

Spatial AI Canvas: An Intelligent Interactive Surface for Real-Time Visual Reasoning

B.Tech Computer Science & Engineering
Semester 8
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PROJECT WORK / PRACTICE SCHOOL

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Problem Statement

1. The "Linear Information Trap" (Cognitive Overload)

Current AI educational tools are primarily chat-centric (e.g., ChatGPT, Gemini). These interfaces force a linear flow of information, which contradicts how the human brain naturally processes complex subjects. In high-level STEM or theoretical learning, information is often non-linear and relational. When a student uses a chatbot, they are forced to scroll through long histories, losing the visual context of previous explanations, leading to high cognitive load and fragmented understanding.

2. Disconnect Between Annotation and Explanation

Current learning tools create a workflow and context gap, forcing students to switch between their notes and separate AI platforms, disrupting their flow and efficiency. Students must manually convert visual thinking (sketches, diagrams, annotations) into text and reapply responses back into their workspace. Additionally, AI lacks spatial awareness, treating queries as isolated inputs rather than understanding a student's position within notes or diagrams, limiting real-time, intuitive learning support.

3. Static vs. Dynamic Visualization

While AI can generate images, these are often static "dead" files that cannot be interacted with. In a learning environment, a diagram of a "Black Hole" or a "Chemical Bond" should be an evolving part of the notes. Current systems lack Real-Time Visual Reasoning, where the AI can adapt a diagram based on a student's specific handwritten question or a request for a language switch (e.g., English to Tamil).

4. Language Barriers in Multimodal Contexts

Educational AI is predominantly English-optimized. When local language support (like Tamil or Hindi) exists, it is usually limited to text translation. There is a lack of multimodal bilingual support where an AI can simultaneously understand a handwritten English term and generate a localized, annotated visual explanation in a regional language without breaking the user's creative flow.

Project Objectives

- **Bridge the Workflow Gap:**
Seamlessly integrate AI intelligence directly into an infinite canvas to eliminate task-switching between notes and chat windows.
- **Implement Spatial Reasoning**
Develop a system that recognizes the coordinates and layout of user sketches to provide context-aware assistance.
- **Enable Multimodal Knowledge Synthesis**
Synthesize real-time explanations, dynamic diagrams, and voice-to-visual responses based on the student's active workspace.
- **Deliver Multilingual Accessibility**
Provide instant, localized content (e.g., English to Hindi or Arabic) for both text and diagrams to support diverse learning needs.
- **Eliminate Manual Data Silos**
Automate the transformation of rough student annotations into structured, AI-enhanced knowledge maps without manual copy-pasting.

Literature Review & Research Gap

1. Generative Chatbots (2022–2024)

Tools like ChatGPT and Gemini established the "Chat Interface" as the standard.

Finding: Research by Hasanbaşoğlu et al. (2026) shows that while LLMs improve brainstorming and drafting, they create a "linear dependency."

Limitation: Information is trapped in a vertical scroll. Students often experience "Cognitive Offloading," where they delegate thinking to the AI rather than engaging with the material (Gerlich, 2025).

2. Multimodal & Vision Integration (2024–2025)

The Shift: Transitioned from text-only models to Vision-Language Models (VLMs), enabling AI to "see," describe diagrams, and caption visual instructional aids.

Key Insight (Microsoft, 2025): AI adoption for accessibility reached 33%, but research proved AI is most effective as "Co-Intelligence"—it performs better when paired with active student note-taking than as a standalone tutor.

The "Static Observer" Gap: While AI could identify objects, it lacked Spatial Grounding. It could not interpret the "spatial logic" of a student's hand-drawn sketches or understand how coordinates on a digital board relate to conceptual ideas.

3. Agentic Spatial Intelligence (2025–2026)

The Shift: Moving from passive "Chatbots" to Agentic AI—systems that don't just talk but observe and act within a 2D environment to co-create with the user.

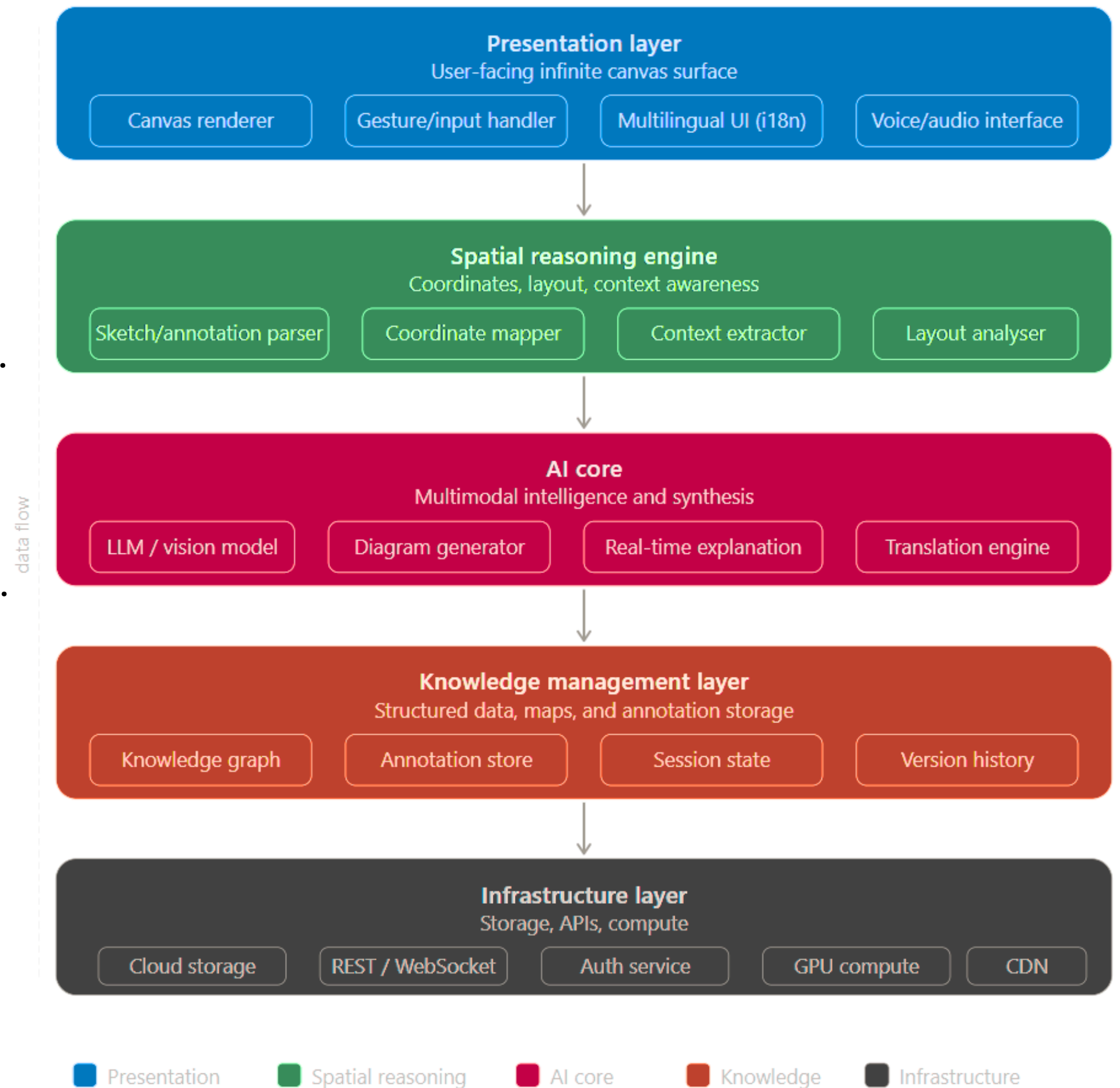
Key Insight (MIT AI Summit, 2025): Recent research identifies "Spatial Grounding" as the next frontier; AI must move beyond describing pixels to understanding the mathematical relationship between coordinates (x, y) on a canvas.

The "Contextual Blindness" Gap: Current models still treat every user input as a standalone string. They lack the long-term spatial memory to recognize that a sketch made ten minutes ago in the top-right corner is the subject of a current question.

System Architecture

Five-layer stack where each layer has a single responsibility and passes data to the one below it.

1. **Presentation Layer** — the infinite canvas surface. Captures touch, stylus, and voice input. Renders all AI outputs back to the user with multilingual support.
2. **Spatial Reasoning Engine** — the perceptual layer. Detects position, shape, and relationships of drawn elements and packages spatial context before any AI call is made.
3. **AI Core** — the intelligence layer. Receives the canvas snapshot and spatial context together, then generates explanations, diagrams, and real-time translations.



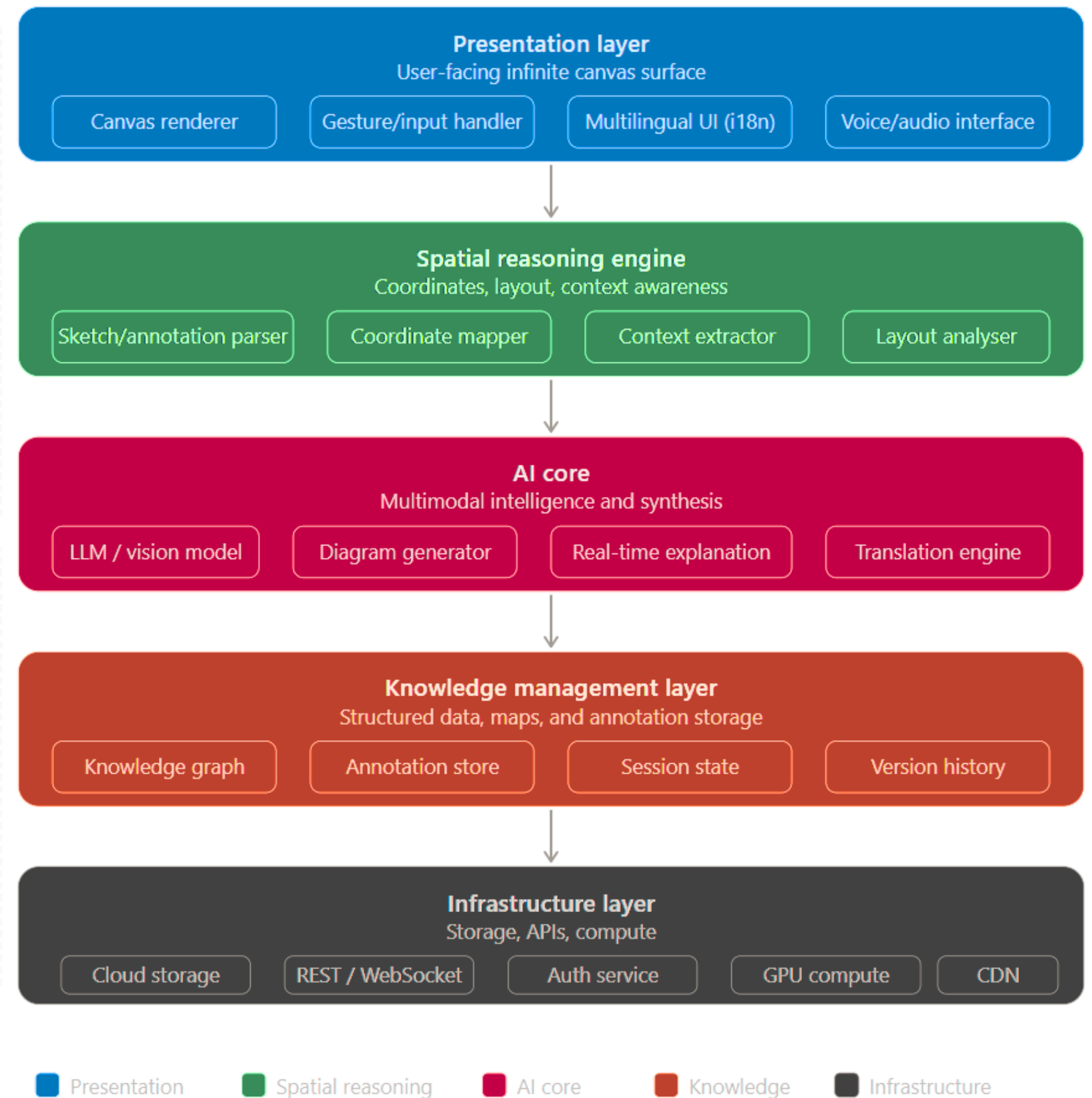
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4. **Knowledge Management Layer** — the memory layer. Converts raw annotations into a structured knowledge graph and maintains session state and version history.

5. **Infrastructure Layer** — the backbone. Handles real-time sync, GPU-accelerated inference, authentication, and global content delivery.

Each layer operates independently yet feeds into a single, tightly integrated pipeline ensuring the system is modular, scalable, and maintainable.



Progress Completed



Planning & Setup — Completed

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- Project idea finalized and objectives defined
- System architecture designed (5-layer stack)
- Technology stack selected and documented
- GitHub repository created and initialized
- Project folder structure set up



Frontend

Frontend — Just Started

- React.js project initialized
- Basic canvas component created
- Simple blank canvas renders on screen

Overall Progress — Early Stage

The initial phase has been dedicated entirely to research, planning, and environment setup. The architecture and tech stack are locked in, and the frontend scaffolding has just begun. Development of core features will kick off in the coming days following the defined pipeline.

Expected Outcomes

→ **A unified AI-powered workspace**

Students will no longer need to switch between note-taking apps and AI chat tools. Everything drawing, querying, and receiving responses happens on a single infinite canvas.

→ **Context-Aware AI Assistance**

An AI system that understands the spatial layout of the canvas and delivers relevant explanations, answers, and suggestions based on exactly what the user has drawn or written.

→ **Multimodal Knowledge Synthesis**

The ability to generate real-time visual diagrams, structured summaries, and voice-to-visual responses directly onto the canvas in response to user queries.

→ **Multilingual Accessibility**

Seamless translation of both text and generated diagrams into multiple languages including Arabic and Hindi, making the tool accessible to diverse student communities.

→ **Automated Knowledge Mapping**

A system that automatically transforms rough student annotations and sketches into clean, structured, AI-enhanced knowledge maps eliminating manual data organization entirely.